SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING **COURSE CODE: CSA4305**

COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS

Weightage: 10%

QUESTIONS: **Total Marks: 10**

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
```

```
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP C. SMTP
D. SNMP
```

Questions:

1. Select the correct HTML tag for video embedding and justify your answer.

Correct Answer: B. <video src="video.mp4" controls></video>

The <video> element is the standard HTML5 tag used to embed video content in a webpage. It allows users to play videos directly in the browser without requiring additional plugins.

Example

```
<video width="640" height="360" controls>
  <source src="video.mp4" type="video/mp4">
  Your browser does not support the video tag.
</video>
```

Justification

- It is an official HTML5 element.
- Supports video playback in all modern browsers.
- The controls attribute displays Play, Pause, Volume, and Full Screen buttons.
- Multiple video formats such as MP4, WebM, and Ogg are supported.
- Improves user experience and accessibility.

Why the other options are incorrect

- **A. <media>** – Not a valid HTML element.
- **C. <movie>** – Not supported in HTML5.
- **D. <vid>** – Not an HTML tag.

Conclusion

The <video> element is the correct choice because it follows HTML5 standards and provides built-in browser support for multimedia content.

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2. Identify the appropriate HTTP method for form submission and explain why.

Correct Answer: B. POST

The **POST** method is used to send user data securely from the browser to the server. Unlike GET, POST sends data in the request body instead of the URL.

Example

```
<form action="/register" method="POST">

<label>Name</label>
<input type="text" name="name">

<label>Email</label>
<input type="email" name="email">

<button type="submit">Register</button>

</form>
```

Why POST is used

- Keeps sensitive information out of the URL.
- Suitable for registration and login forms.
- Supports large amounts of data.
- Works securely with HTTPS.
- Prevents sensitive data from appearing in browser history.

Why other options are incorrect

- **GET** – Data appears in the URL and is not suitable for confidential information.
- **PUT** – Mainly used to update existing resources in REST APIs.
- **DELETE** – Used to delete resources from the server.

Conclusion

POST is the most appropriate HTTP method for form submission because it provides better security, privacy, and reliability for transmitting user information.

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3. Choose the proper semantic element for navigation and justify its usage.

Correct Answer: B. <nav>

The <nav> element is a semantic HTML5 tag specifically designed to contain navigation links.

Example

```
<nav>

<a href="index.html">Home</a>

<a href="about.html">About</a>

<a href="services.html">Services</a>

<a href="contact.html">Contact</a>

</nav>
```

Justification

- Clearly identifies the navigation section.
- Improves webpage structure.
- Helps search engines understand the website.
- Improves accessibility for screen readers.
- Makes the code easier to maintain.

Why other options are incorrect

- **<div>** – Generic container with no semantic meaning.
- **<section>** – Used for grouping related content, not specifically navigation.
- **<header>** – Represents introductory content and may contain navigation, but it is not the navigation element itself.

Conclusion

The <nav> element should always be used for menus and navigation links because it provides semantic meaning and improves accessibility and SEO.

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4. Identify the correct XML syntax and explain the error in other options.

Correct Answer: B.

```
<note>
```

<to>User</to>
<from>Admin</from>
</note>

This XML document is well-formed because:

- Every opening tag has a matching closing tag.
- Elements are properly nested.
- The root element is closed correctly.
- XML syntax rules are followed.

Why other options are incorrect

Option A

<note><to>User</to><from>Admin</from>

- Missing closing </note> tag.

Option C

<note><to>User</from></note>

- Opening tag <to> is incorrectly closed with </from>.
- Tags must match exactly.

Option D

<note to="User"></note

- Missing closing angle bracket (>).
- Invalid XML syntax.

Importance of Correct XML

- XML files are used for configuration and data exchange.
- Incorrect syntax prevents XML parsers from reading the file.
- Well-formed XML improves reliability and interoperability.

Conclusion

Option **B** is the correct XML syntax because it follows all XML rules, including proper nesting and matching opening and closing tags.

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5. Choose the correct protocol for web communication and justify your choice.

Correct Answer: B. HTTP

HTTP (HyperText Transfer Protocol) is the standard protocol used for communication between web browsers and web servers.

How HTTP Works

1. The user enters a website URL.
2. The browser sends an HTTP request to the server.
3. The server processes the request.

4. The server sends an HTTP response.
5. The browser displays the webpage.

Advantages of HTTP

- Enables communication between client and server.
- Supports web pages, images, videos, and APIs.
- Works with HTML, CSS, JavaScript, and XML.
- Simple and widely supported.
- Can be secured using HTTPS.

Why other options are incorrect

- **FTP (File Transfer Protocol)** – Used for transferring files, not web page communication.
- **SMTP (Simple Mail Transfer Protocol)** – Used for sending emails.
- **SNMP (Simple Network Management Protocol)** – Used for network monitoring and management.

Conclusion

HTTP is the correct protocol for web communication because it enables browsers and web servers to exchange web resources efficiently. For secure communication, HTTP is commonly used with SSL/TLS as **HTTPS**, which encrypts the transmitted data and protects user information.

Code of Conduct:

I Mathusri P-192311363 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: *Mathusri P*

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, wellstructured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand